```
# Step 3: Vectorize Text
vectorizer = CountVectorizer(stop_words='english') X_train_vectorized =
vectorizer.fit_transform(X_train) X_test_vectorized = vectorizer.transform(X_test) # Step 4: Train
Model
model = MultinomialNB()
model.fit(X_train_vectorized, y_train) # Step 5: Evaluate Model
predictions = model.predict(X_test_vectorized) print(classification_report(y_test, predictions))
print(f'Accuracy: {accuracy_score(y_test, predictions)}') # Step 6: Real-Time Prediction
new_review = ["The product was excellent and exceeded expectations."]
new_review_vectorized = vectorizer.transform(new_review) print(f"Predicted Sentiment:
{model.predict(new_review_vectorized)[0]}")
```

Project 4: Image Classification with CNNs

Objective:

To classify images into predefined categories using Convolutional Neural Networks (CNNs). This project demonstrates the power of deep learning in various domains, including quality control, healthcare, and retail.

Detailed Plan:

1. Dataset Selection:
    - Use publicly available datasets like CIFAR10 (for small object categories) or ImageNet (for large-scale image classification).
    - Split the dataset into training, validation, and test sets to ensure robust evaluation.
    - Optionally, collect domain-specific data and annotate it using tools like LabelImg for custom applications.

2. Data Preprocessing:
    - Resize images to a consistent input size (e.g., 32x32 for CIFAR10 or 224x224 for ImageNet).
    - Normalize pixel values to a range of [0, 1] or [-1, 1].
    - Apply data augmentation techniques such as flipping, rotation, scaling, and cropping to enhance generalization.

3. Model Building: